

VINAYA AYINAMPUDI

vinchow9999@gmail.com | LinkedIn | GitHub | Riverside, CA

EDUCATION

University of California, Berkeley | Berkeley, CA
B.A. in Data Science + Computer Science | Expected June 2030

TECHNICAL SKILLS

Languages: Python, Java, SQL | **ML/Data Science:** TensorFlow, CNNs, NumPy, Pandas,
OpenCV Hardware: Raspberry Pi, Arduino, Fusion 360, CNC Milling, 3D Printing
Developer Tools: Git, GitHub, VS Code, Excel

EXPERIENCE

Youth STEAM Advisors | Founder | 2022-Present

- Founded and scaled an international STEM nonprofit with 1000+ student members and 150+ directors and ambassadors
- Designed and led month long workshops, technical blogs, and STEM educational resources for younger students
- Partnered with public libraries to coordinate hands on technical workshops introducing students to engineering and computational thinking

World Robot Olympiad (WRO) | Team Captain & Competitor | 2023-2024

- Engineered autonomous robotic vehicles using TensorFlow, Raspberry Pi, CNN based computer vision, and real time inference pipelines for robotics navigation challenges
- Won 1st Place at WRO Nationals in Future Engineers and Future Innovators divisions and represented Team USA internationally in Panama and Türkiye
- Implemented machine learning workflows for autonomous path planning, obstacle navigation, and robotic decision making under competition constraints

UCSD ABLE Research Program | Summer Research Intern | Summer 2024

- Selected as 1 of fewer than 15 students nationwide for a fully funded machine learning and biomedical engineering research experience
- Studied probability, statistics, linear algebra, perceptrons, machine learning algorithms, and data science applications at the San Diego Supercomputer Center
- Applied quantitative and computational techniques to technical research problems involving biomedical datasets

Science & Engineering Research | ML Researcher | 2023-2025

- Developed machine learning and deep learning models to compare classification accuracy for Down Syndrome detection using pediatric facial imaging datasets
- Built a deep convolutional neural network utilizing 9 prenatal ultrasound markers to automate Down Syndrome risk detection
- Presented research to 14+ judges and advanced twice to the California Science & Engineering Fair after earning 1st Place in county competition